The modal-Gamma-distribution

Parametrisation

The modal-Gamma-distribution has the following density

$$\pi(y) = \frac{b^a}{\Gamma(a)} y^{a-1} \exp(-by), \qquad a > 1, \quad b > 0, \quad y > 0.$$

where the mode $m = (a-1)/b$ is the target of the regression model instead of the mean or quantile. **Note the constraint** $a > 1$.

With precision

$$\tau = (s\phi)/m^2$$

for fixed scaling s and precision parameter ϕ, this complete the spesification. The mapping back to the (a, b)-parameterisation is then $\phi' = s\phi$, $\delta = \left(\sqrt{\phi'(\phi'+4)} + \phi'\right)/2$, then

$$a = 1 + \delta \quad \text{and} \quad b = \delta/m$$

Link-function

The linear predictor η is linked to the mode m using a default log-link

$$m = \exp(\eta)$$

Hyperparameter

The hyperparameter is the precision parameter ϕ, which is represented as

$$\phi = \exp(\theta)$$

and the prior is defined on θ.

Specification

- `family="mgamma"` for regression models and `family="mgamma.surv"` for survival models.
- Required arguments: for `mgamma.surv`, y (to be given in a format by using `inla.surv()`), and for `mgamma`, y and s (default value 1).

The scalings s is **not** used for `family="mgamma.surv"`.

Hyperparameter spesification and default values

doc `The modal Gamma likelihood`

hyper

theta

hyperid `58002`
name `precision parameter`
short.name `prec`
output.name `Precision-parameter for the modal Gamma observations`
output.name.intern `Intern precision-parameter for the modal Gamma observations`
initial `4.60517018598809`

fixed FALSE
prior loggamma
param 1 0.01
to.theta function(x) log(x)
from.theta function(x) exp(x)

survival FALSE

discrete FALSE

link default log

pdf mgamma

doc The modal Gamma likelihood (survival)

hyper

theta1
hyperid 58121
name precision parameter
short.name prec
output.name Precision-parameter for the modal Gamma surv observations
output.name.intern Intern precision-parameter for the modal Gamma surv observations
initial 0
fixed FALSE
prior loggamma
param 1 0.01
to.theta function(x) log(x)
from.theta function(x) exp(x)

theta2
hyperid 58122
name beta1
short.name beta1
output.name beta1 for modal Gamma-Cure
output.name.intern beta1 for modal Gamma-Cure
initial -7
fixed FALSE
prior normal
param -4 100
to.theta function(x) x
from.theta function(x) x

theta3
hyperid 58123
name beta2
short.name beta2
output.name beta2 for modal Gamma-Cure
output.name.intern beta2 for modal Gamma-Cure

initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 58124
name beta3
short.name beta3
output.name beta3 for modal Gamma-Cure
output.name.intern beta3 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 58125
name beta4
short.name beta4
output.name beta4 for Ga mma-Cure
output.name.intern beta4 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 58126
name beta5
short.name beta5
output.name beta5 for modal Gamma-Cure
output.name.intern beta5 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 58127

name beta6
short.name beta6
output.name beta6 for modal Gamma-Cure
output.name.intern beta6 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 58128
name beta7
short.name beta7
output.name beta7 for modal Gamma-Cure
output.name.intern beta7 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta9

hyperid 58129
name beta8
short.name beta8
output.name beta8 for modal Gamma-Cure
output.name.intern beta8 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta10

hyperid 58130
name beta9
short.name beta9
output.name beta9 for modal Gamma-Cure
output.name.intern beta9 for modal Gamma-Cure
initial 0
fixed FALSE
prior normal
param 0 100

to.theta `function(x) x`

from.theta `function(x) x`

theta11

hyperid `58131`

name `beta10`

short.name `beta10`

output.name `beta10 for modal Gamma-Cure`

output.name.intern `beta10 for modal Gamma-Cure`

initial `0`

fixed `FALSE`

prior `normal`

param `0 100`

to.theta `function(x) x`

from.theta `function(x) x`

survival `TRUE`

discrete `FALSE`

link `default log neglog`

pdf `agamma`

Example

In the following example we estimate the parameters in a simulated example.

```
n <- 10^4
phi <- 2.2
x <- rnorm(n)
m <- exp(1.1 + 1.2 * x)
delta <- (sqrt(phi * (phi + 4.0)) + phi) / 2.0

y <- rgamma(n, shape = 1 + delta, rate = delta / m)
r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "mgamma",
          control.compute = list(cpo = T),
          verbose = TRUE)

Y <- inla.surv(y, 1)
rr <- inla(Y ~ 1 + x,
          data = list(Y = Y, x = x),
          family = "mgammasurv",
          control.compute = list(cpo = T),
          verbose = TRUE)
```

Notes

None.